```
step 1 — Install Libraries
```

```
# Install/Upgrade Hugging Face Transformers and PyTorch (needed to run pretrained models)
!pip install --upgrade transformers torch sentencepiece
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.3.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (0.2.1)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0
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Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.25.1)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
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Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.3.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1
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Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (8.3.1)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (1.5
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Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (0
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Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->h
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->t
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3
```

step 2 — Import Libraries

```
# Import necessary components for direct model loading
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# We'll use this to display outputs neatly (though not strictly needed for this fix)
import pandas as pd
```

step 3 — Prepare Your 15 Sentences (Dataset)

```
# Our input dataset — 15 English sentences of different types
sentences = [
    # Simple sentences
    "The sun rises in the east.",
    "She drinks coffee every morning.",
    "Birds can fly.",

    # Compound sentences
    "I wanted to go outside, but it started raining.",
    "She studied hard, so she passed the exam.",

    # Complex sentences
    "Although he was tired, he continued working.",
    "Because the traffic was heavy, we arrived late.",

    # Interrogative sentences
    "What is the capital of France?",
    "How does machine translation work?",

    # Technical sentences
    "Neural networks use backpropagation to update weights.",
    "Transformers rely on self-attention mechanisms.",

    # Conversational sentences
    "Hey, how are you doing today?",
    "Can you help me with this problem?",
```

```
    # News/Academic style
    "The government announced new policies on climate change.",
    "Researchers discovered a new species of deep-sea fish."
]

print(f"Total sentences: {len(sentences)}")  # Should print 15
```

```
Total sentences: 15
```

step 4 — Load English → French Model & Translate

```
# Define the model name
model_name = "Helsinki-NLP/opus-mt-en-fr"

# Load the tokenizer and model directly
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

# Translate all 15 sentences
french_translations = []

for sentence in sentences:
    # Tokenize the input sentence
    inputs = tokenizer(sentence, return_tensors="pt", padding=True)

    # Generate translation
    outputs = model.generate(**inputs)

    # Decode the generated tokens to text
    translated = tokenizer.decode(outputs[0], skip_special_tokens=True)
    french_translations.append(translated)
    print(f"EN: {sentence}")
    print(f"FR: {translated}\n")
```

```
/usr/local/lib/python3.12/dist-packages/transformers/models/marian/tokenization_marian.py:176: UserWarning: Recommended: pip
  warnings.warn("Recommended: pip install sacremoses.")
```

Loading weights: 100% 258/258 [00:00<00:00, 3550.88it/s]

```
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight
EN: The sun rises in the east.
FR: Le soleil se lève à l'est.

EN: She drinks coffee every morning.
FR: Elle boit du café tous les matins.

EN: Birds can fly.
FR: Les oiseaux peuvent voler.

EN: I wanted to go outside, but it started raining.
FR: Je voulais sortir, mais il a commencé à pleuvoir.

EN: She studied hard, so she passed the exam.
FR: Elle a étudié dur, donc elle a réussi l'examen.

EN: Although he was tired, he continued working.
FR: Même s'il était fatigué, il a continué à travailler.

EN: Because the traffic was heavy, we arrived late.
FR: Parce que le trafic était lourd, nous sommes arrivés en retard.

EN: What is the capital of France?
FR: Quelle est la capitale de la France ?

EN: How does machine translation work?
FR: Comment fonctionne la traduction automatique?

EN: Neural networks use backpropagation to update weights.
FR: Les réseaux neuraux utilisent la rétropropagation pour mettre à jour les poids.

EN: Transformers rely on self-attention mechanisms.
FR: Les transformateurs s'appuient sur des mécanismes d'auto-attention.

EN: Hey, how are you doing today?
FR: Comment ça va aujourd'hui ?

EN: Can you help me with this problem?
FR: Pouvez-vous m'aider avec ce problème ?

EN: The government announced new policies on climate change.
FR: Le gouvernement a annoncé de nouvelles politiques sur le changement climatique.

EN: Researchers discovered a new species of deep-sea fish.
FR: Les chercheurs ont découvert une nouvelle espèce de poissons d'eau profonde.
```

step 5 — Load English → German Model & Translate

```
# Define the model name for English to German
model_name_de = "Helsinki-NLP/opus-mt-en-de"

# Load the tokenizer and model directly for German
tokenizer_de = AutoTokenizer.from_pretrained(model_name_de)
model_de = AutoModelForSeq2SeqLM.from_pretrained(model_name_de)

german_translations = []

for sentence in sentences:
    # Tokenize the input sentence
    inputs = tokenizer_de(sentence, return_tensors="pt", padding=True)

    # Generate translation
    outputs = model_de.generate(**inputs)

    # Decode the generated tokens to text
    translated = tokenizer_de.decode(outputs[0], skip_special_tokens=True)
    german_translations.append(translated)
    print(f"EN: {sentence}")
    print(f"DE: {translated}\n")
```

```
tokenizer_config.json: 100%                                    42.0/42.0 [00:00<00:00, 3.42kB/s]

source.spm: 100%                                               768k/768k [00:00<00:00, 20.8MB/s]

target.spm: 100%                                               797k/797k [00:00<00:00, 38.1MB/s]

vocab.json:        1.27M/? [00:00<00:00, 30.7MB/s]

pytorch_model.bin: 100%                                        298M/298M [00:03<00:00, 150MB/s]

Loading weights: 100%                                          258/258 [00:00<00:00, 6173.72it/s]

model.safetensors: 100%                                        298M/298M [00:04<00:00, 108MB/s]
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weig
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weig
generation_config.json: 100%                                   293/293 [00:00<00:00, 21.0kB/s]
EN: The sun rises in the east.
DE: Die Sonne geht im Osten auf.

EN: She drinks coffee every morning.
DE: Sie trinkt jeden Morgen Kaffee.

EN: Birds can fly.
DE: Vögel können fliegen.

EN: I wanted to go outside, but it started raining.
DE: Ich wollte rausgehen, aber es fing an zu regnen.

EN: She studied hard, so she passed the exam.
DE: Sie hat hart studiert, also hat sie die Prüfung bestanden.

EN: Although he was tired, he continued working.
DE: Obwohl er müde war, arbeitete er weiter.

EN: Because the traffic was heavy, we arrived late.
DE: Weil der Verkehr schwer war, kamen wir spät an.

EN: What is the capital of France?
DE: Was ist die Hauptstadt Frankreichs?

EN: How does machine translation work?
DE: Wie funktioniert die maschinelle Übersetzung?

EN: Neural networks use backpropagation to update weights.
DE: Neurale Netzwerke verwenden Backpropagation, um Gewichte zu aktualisieren.

EN: Transformers rely on self-attention mechanisms.
DE: Transformer setzen auf Selbstaufmerksamkeitsmechanismen.

EN: Hey, how are you doing today?
DE: Hey, wie geht's dir heute?

EN: Can you help me with this problem?
DE: Können Sie mir bei diesem Problem helfen?

EN: The government announced new policies on climate change.
DE: Die Regierung kündigte eine neue Klimapolitik an.

EN: Researchers discovered a new species of deep-sea fish.
DE: Forscher entdeckten eine neue Art von Tiefseefischen.
```

step 6 — Load English → Hindi Model & Translate

```
# Define the model name for English to Hindi
model_name_hi = "Helsinki-NLP/opus-mt-en-hi"

# Load the tokenizer and model directly for Hindi
tokenizer_hi = AutoTokenizer.from_pretrained(model_name_hi)
model_hi = AutoModelForSeq2SeqLM.from_pretrained(model_name_hi)

hindi_translations = []

for sentence in sentences:
    # Tokenize the input sentence
    inputs = tokenizer_hi(sentence, return_tensors="pt", padding=True)

    # Generate translation
    outputs = model_hi.generate(**inputs)

    # Decode the generated tokens to text
    translated = tokenizer_hi.decode(outputs[0], skip_special_tokens=True)
    hindi_translations.append(translated)
    print(f"EN: {sentence}")
```

```
    print(f"HI: {translated}\n")
```

```
tokenizer_config.json: 100%  44.0/44.0 [00:00<00:00, 3.54kB/s]

source.spm: 100%  812k/812k [00:00<00:00, 22.8MB/s]

target.spm: 100%  1.07M/1.07M [00:00<00:00, 47.2MB/s]

vocab.json:  2.10M/? [00:00<00:00, 41.7MB/s]

/usr/local/lib/python3.12/dist-packages/transformers/models/marian/tokenization_marian.py:176: UserWarning: Recommended: p
  warnings.warn("Recommended: pip install sacremoses.")

pytorch_model.bin: 100%  306M/306M [00:03<00:00, 173MB/s]

Loading weights: 100%  258/258 [00:00<00:00, 6250.68it/s]

model.safetensors: 100%  306M/306M [00:03<00:00, 132MB/s]

The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weig
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weig

generation_config.json: 100%  293/293 [00:00<00:00, 7.18kB/s]

EN: The sun rises in the east.
HI: सूरज पूर्व में उदय होता है.

EN: She drinks coffee every morning.
HI: वह कॉफी हर सुबह पी रही है.

EN: Birds can fly.
HI: पक्षी उड़ सकते हैं.

EN: I wanted to go outside, but it started raining.
HI: मैं बाहर जाना चाहता था, लेकिन यह बारिश शुरू कर दिया.

EN: She studied hard, so she passed the exam.
HI: वह कठिन अध्ययन करती थी, सो वह परीक्षा से गुज़रती थी ।

EN: Although he was tired, he continued working.
HI: हालाँकि वह थक गया था, फिर भी वह काम करता रहा ।

EN: Because the traffic was heavy, we arrived late.
HI: क्योंकि यातायात भारी था, हम देर से पहुँच गए ।

EN: What is the capital of France?
HI: फ्रांस की राजधानी क्या है?

EN: How does machine translation work?
HI: मशीन अनुवाद कैसे काम करता है?

EN: Neural networks use backpropagation to update weights.
HI: तंत्रिका नेटवर्क वज़न अद्यतन करने के लिए पीठ-शोषण का उपयोग करते हैं.

EN: Transformers rely on self-attention mechanisms.
HI: कायापलट करनेवाले खुद पर भरोसा रखते हैं।

EN: Hey, how are you doing today?
HI: अरे, आप आज कैसे कर रहे हैं?

EN: Can you help me with this problem?
HI: आप इस समस्या के साथ मेरी मदद कर सकते हैं?

EN: The government announced new policies on climate change.
HI: सरकार ने जलवायु परिवर्तन पर नए नियमों की घोषणा की.

EN: Researchers discovered a new species of deep-sea fish.
HI: खोजकर्ताओं ने गहरी समुद्री मछलियों की एक नई प्रजाति खोज की.
```

step 7 — Compare All Translations in a Table

```
# Create a comparison DataFrame to view all translations side by side
df = pd.DataFrame({
    "English": sentences,
    "French": french_translations,
    "German": german_translations,
    "Hindi": hindi_translations
})

# Display the table
df
```

	English	French	German	Hindi
0	The sun rises in the east.	Le soleil se lève à l'est.	Die Sonne geht im Osten auf.	सूरज पूर्व में उदय होता है.
1	She drinks coffee every morning.	Elle boit du café tous les matins.	Sie trinkt jeden Morgen Kaffee.	वह कॉफी हर सुबह पी रही है.
2	Birds can fly.	Les oiseaux peuvent voler.	Vögel können fliegen.	पक्षी उड़ सकते हैं.
3	I wanted to go outside, but it started raining.	Je voulais sortir, mais il a commencé à pleuvoir.	Ich wollte rausgehen, aber es fing an zu regnen.	मैं बाहर जाना चाहता था, लेकिन यह बारिश शुरू कर...
4	She studied hard, so she passed the exam.	Elle a étudié dur, donc elle a réussi l'examen.	Sie hat hart studiert, also hat sie die Prüfun...	वह कठिन अध्ययन करती थी, सो वह परीक्षा से गुज़र...
5	Although he was tired, he continued working.	Même s'il était fatigué, il a continué à trava...	Obwohl er müde war, arbeitete er weiter.	हालाँकि वह थक गया था, फिर भी वह काम करता रहा।
6	Because the traffic was heavy, we arrived late.	Parce que le trafic était lourd, nous sommes a...	Weil der Verkehr schwer war, kamen wir spät an.	क्योंकि यातायात भारी था, हम देर से पहुँच गए।
7	What is the capital of France?	Quelle est la capitale de la France ?	Was ist die Hauptstadt Frankreichs?	फ्रांस की राजधानी क्या है?
8	How does machine translation work?	Comment fonctionne la traduction automatique?	Wie funktioniert die maschinelle Übersetzung?	मशीन अनुवाद कैसे काम करता है?
9	Neural networks use backpropagation to update ...	Les réseaux neuraux utilisent la rétropropagat...	Neurale Netzwerke verwenden Backpropagation, u...	तंत्रिका नेटवर्क वज़न अद्यतन करने के लिए पीठ-श...
10	Transformers rely on self-attention mechanisms.	Les transformateurs s'appuient sur des mécanis...	Transformer setzen auf Selbstaufmerksamkeitsme...	कायापलट करनेवाले खुद पर भरोसा रखते हैं।
11	Hey, how are you doing today?	Comment ça va aujourd'hui ?	Hey, wie geht's dir heute?	अरे, आप आज कैसे कर रहे हैं?
12	Can you help me with this problem?	Pouvez-vous m'aider avec ce problème ?	Können Sie mir bei diesem Problem helfen?	आप इस समस्या के साथ मेरी मदद कर सकते हैं?
13	The government announced new policies on clima...	Le gouvernement a annoncé de nouvelles politiq...	Die Regierung kündigte eine neue Klimapolitik an.	सरकार ने जलवायु परिवर्तन पर नए नियमों की घोषणा...

Next steps: Generate code with df New interactive sheet

step 8 — Test with Complex / Long Sentences Specifically

```
# Picking out the complex/technical sentences for deeper analysis
complex_sentences = [
    "Although he was tired, he continued working.",
    "Neural networks use backpropagation to update weights.",
    "Researchers discovered a new species of deep-sea fish.",
    "Because the traffic was heavy, we arrived late.",
    "Transformers rely on self-attention mechanisms."
]

print("=== Complex Sentence Translation Test (English → French) ===\n")

for s in complex_sentences:
    # Tokenize the input sentence using the French tokenizer (tokenizer defined in xqUyIez6eSs8)
    inputs = tokenizer(s, return_tensors="pt", padding=True)

    # Generate translation using the French model (model defined in xqUyIez6eSs8)
    outputs = model.generate(**inputs)

    # Decode the generated tokens to text
    translated = tokenizer.decode(outputs[0], skip_special_tokens=True)

    print(f"ORIGINAL : {s}")
    print(f"FRENCH   : {translated}")
    print("-" * 60)
```

```
=== Complex Sentence Translation Test (English → French) ===

ORIGINAL : Although he was tired, he continued working.
FRENCH   : Même s'il était fatigué, il a continué à travailler.
------------------------------------------------------------
ORIGINAL : Neural networks use backpropagation to update weights.
FRENCH   : Les réseaux neuraux utilisent la rétropropagation pour mettre à jour les poids.
------------------------------------------------------------
ORIGINAL : Researchers discovered a new species of deep-sea fish.
FRENCH   : Les chercheurs ont découvert une nouvelle espèce de poissons d'eau profonde.
------------------------------------------------------------
ORIGINAL : Because the traffic was heavy, we arrived late.
FRENCH   : Parce que le trafic était lourd, nous sommes arrivés en retard.
------------------------------------------------------------
ORIGINAL : Transformers rely on self-attention mechanisms.
FRENCH   : Les transformateurs s'appuient sur des mécanismes d'auto-attention.
```